Week 1

Lecture 1 Introduction to vector spaces

This covers part of ***Section 1.1*** *in the course notes.*

In three-dimensional geometry, a vector is defined geometrically by three coordinates: x, y, and z. A vector is often represented as an (ordered) triple

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a & b & c \end{bmatrix}^T$$

where $a, b, c \in \mathbb{R}$. Here a represents the value of the x-coordinate, b the y-coordinate, and c the z-coordinate. We denote by $\mathbb{R}^3$ the set of all such triples.

As you will have seen in MATH 136, one can immediately generalize this in the following two ways:

- Instead of working with three coordinates x, y, z, we work with $n \geq 1$ coordinates $x_1, x_2, ..., x_n$. Then a vector $\vec{v}$ is represented as an (ordered) n-tuple

$$\vec{v} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}^T$$

  for $a_1, ..., a_n \in \mathbb{R}$. We denote by $\mathbb{R}^n$ the set of all such n-tuples.

- Instead of asking for our tuple values to be in $\mathbb{R}$, we can also ask for them to be in $\mathbb{C}$. Thus, a vector here is an n-tuple:

$$\vec{v} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}^T$$

  for $a_1, ..., a_n \in \mathbb{C}$. We denote by $\mathbb{C}^n$ the set of all such n-tuples.

Let us use $\mathbb{F}$ to denote either $\mathbb{R}$ or $\mathbb{C}$. Three fundamental operations carried out on $\mathbb{F}^n$ are addition, scalar multiplication, and the dot product. These are defined as follows: given two vectors

$$\vec{v}_1 = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

in $\mathbb{F}^n$, and a scalar $\alpha \in \mathbb{F}$, we define the addition of $\vec{v}_1$ with $\vec{v}_2$ by:

$$\vec{v}_1 + \vec{v}_2 = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{bmatrix}$$

scalar multiplication of α with $\vec{v}_1$ by:

$$\alpha \cdot \vec{v}_1 = \alpha \cdot \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} \alpha a_1 \\ \vdots \\ \alpha a_n \end{bmatrix}$$

and the dot product (or standard inner product) of $\vec{v}_1$ with $\vec{v}_2$ by:

$$\vec{v}_1 \cdot \vec{v}_2 = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = a_1\overline{b_1} + \cdots + a_n\overline{b_n}$$

In MATH 235, we study a further generalization of the above, in which vectors are treated as abstract objects in a set equipped with suitably-defined addition and scalar multiplication, as well as a dot product operation (if desired).

For the first half of the course we will restrict ourselves to assuming our set only has addition and scalar multiplication. As we shall see, this means we will not yet have a full geometric intuition of vectors (at least to the extent we do with $\mathbb{R}^n$ or $\mathbb{C}^n$). Notably, we will not have the notion of *length* of a vector or that of an *angle* between two vectors. But we will be able to derive a lot of the theory just from our two operations.

In the second half of the course, we will introduce a generalization of the dot product on our set, which will then enable us to define notions such as *length*, *angle*, and *orthogonality* for vectors in the abstract.

Throughout the course, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

Definition 1.1 (Vector space)

A **vector space** over $\mathbb{F}$ is a set V equipped with two operations $+ : V \times V \to V$ (called addition) and $\cdot : \mathbb{F} \times V \to V$ (called scalar multiplication) which satisfy the following axioms:

[A1] For all $\vec{u}, \vec{v}, \vec{w} \in V$, $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$.

[A2] There exists an element $\vec{0} \in V$ such that for all $\vec{v} \in V$, $\vec{v} + \vec{0} = \vec{0} + \vec{v} = \vec{v}$. $\vec{0}$ is unique and is called the **zero vector**.

[A3] For all $\vec{v} \in V$, there exists an element $-\vec{v} \in V$ such that $\vec{v} + (-\vec{v}) = (-\vec{v}) + \vec{v} = \vec{0}$. $-\vec{v}$ is unique and is called the **negation** of $\vec{v}$.

[A4] For all $\vec{u}, \vec{v} \in V$, $\vec{u} + \vec{v} = \vec{v} + \vec{u}$.

[M1] For all $\vec{v} \in V$ and $\alpha, \beta \in \mathbb{F}$, $\alpha \cdot (\beta \cdot \vec{v}) = (\alpha\beta) \cdot \vec{v}$

[M2] For all $\vec{v} \in V$, $1 \cdot \vec{v} = \vec{v}$.

[D1] For all $\vec{u}, \vec{v} \in V$ and $\alpha \in \mathbb{F}$, $\alpha \cdot (\vec{u} + \vec{v}) = \alpha \cdot \vec{u} + \alpha \cdot \vec{v}$.

[D2] For all $\vec{v} \in V$ and $\alpha, \beta \in \mathbb{F}$, $(\alpha + \beta) \cdot \vec{v} = \alpha \cdot \vec{v} + \beta \cdot \vec{v}$.

The elements of V are called **vectors** and the elements of $\mathbb{F}$ which act on the elements of V are called **scalars**. We will often refer to the axioms above as the *vector space axioms*.

There is quite a bit to say about the definition we just gave.

Remark 1.2

(i) It is common and convenient to omit the $\cdot$ in expressions involving scalar multiplication (as long as it doesn't lead to any ambiguity). For instance, $\alpha \cdot \vec{v}$ can be written as $\alpha\vec{v}$.

It is also convenient to write $+(-\vec{v})$ as just $-\vec{v}$ in expressions where a negation is being added. For instance, the expression $\vec{u} + (-\vec{v})$ can be written as $\vec{u} - \vec{v}$. Notice that this gives us another operation on V we can define: $- : V \times V \to V$ given by:

$$\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$$

for all $\vec{u}, \vec{v} \in V$. We call this the **subtraction** operation on V.

(ii) Property **[A1]** allows us to write an expression involving the sum of three vectors without parentheses. We shall often write $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$ simply as $\vec{u} + \vec{v} + \vec{w}$ because the order of operations won't matter. This generalizes to a sum of four or more vectors as well.

Similarly **[M1]** tells us that $\alpha(\beta\vec{v}) = (\alpha\beta)\vec{v}$ can be written as $\alpha\beta\vec{v}$ if we wish.

(iii) Let us clarify what we mean by uniqueness in **[A2]** and **[A3]**. For **[A2]**, we mean that if there was another element $\vec{z} \in V$ which also has the property that $\vec{v} + \vec{z} = \vec{z} + \vec{v} = \vec{v}$ for all $\vec{v} \in V$, then we must have $\vec{z} = \vec{0}$ (check this!)

For **[A3]**, we mean that for any $\vec{v} \in V$, if there was another element $\vec{w} \in V$ which also has the property that $\vec{v} + \vec{w} = \vec{w} + \vec{v} = \vec{0}$, then we must have $\vec{w} = -\vec{v}$ (check this too!)

Example 1.3 (Space of n-tuples $\mathbb{F}^n$)

The set $\mathbb{F}^n$ with the usual coordinate-wise addition and scalar multiplication, as described above, makes it a vector space over $\mathbb{F}$. The zero vector is simply $\vec{0} = \begin{bmatrix} 0 & \cdots & 0 \end{bmatrix}^T$.

It is a good idea to get a feel for how to verify the vector space axioms. As an example, let us check that some of them hold for $\mathbb{F}^2$. Let:

$$\vec{u} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Also let $\alpha, \beta \in \mathbb{F}$.

[A1] We see that:

$$\begin{aligned}\vec{u} + (\vec{v} + \vec{w}) &= \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \right) \\ &= \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_1 + c_1 \\ b_2 + c_2 \end{bmatrix} \\ &= \begin{bmatrix} a_1 + (b_1 + c_1) \\ a_2 + (b_2 + c_2) \end{bmatrix} \\ &= \begin{bmatrix} a_1 + b_1 + c_1 & a_2 + b_2 + c_2 \end{bmatrix}\end{aligned}$$

By doing the same thing with $(\vec{u} + \vec{v}) + \vec{w}$, we get:

$$(\vec{u} + \vec{v}) + \vec{w} = \begin{bmatrix} a_1 + b_1 + c_1 \\ a_2 + b_2 + c_2 \end{bmatrix} = \vec{u} + (\vec{v} + \vec{w})$$

and thus, **[A1]** indeed holds.

[M1] We see that:

$$\begin{aligned}\alpha(\beta\vec{v}) &= \alpha \left(\beta \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right) \\ &= \alpha \begin{bmatrix} \beta b_1 \\ \beta b_2 \end{bmatrix} \\ &= \begin{bmatrix} \alpha\beta b_1 \\ \alpha\beta b_2 \end{bmatrix} \\ &= (\alpha\beta) \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ &= (\alpha\beta)\vec{v}\end{aligned}$$

so **[M1]** holds as well.

Convince yourself that the other six vector space axioms hold for $\mathbb{F}^2$, and moreover that all eight hold for $\mathbb{F}^n$ for any $n \geq 1$. At the very least, this may serve as a good "brush-up" on MATH 136.

Remark 1.4

Notice that $\mathbb{C}^n$ is a vector space over $\mathbb{R}$ as well. All we need to do is restrict the set of scalars from $\mathbb{C}$ to $\mathbb{R}$.

But **warning**: unless otherwise stated, you should always assume that by $\mathbb{F}^n$ we mean a vector space over $\mathbb{F}$ (that is, $\mathbb{R}^n$ is always assumed to be a vector space over $\mathbb{R}$ and $\mathbb{C}^n$ is always assumed to be a vector space over $\mathbb{C}$).

Matrices from MATH 136 make a comeback, and turn out to have a vector space structure too:

Example 1.5 (Space of matrices)

Let $m, n \geq 1$ and consider the set of all $m \times n$ matrices with entries in $\mathbb{F}$:

$$\mathcal{M}_{m\times n}(\mathbb{F}) = \left\{ \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix} : a_{i,j} \in \mathbb{F} \text{ for all } 1 \leq i \leq m, 1 \leq j \leq n \right\}$$

We have the usual entry-wise addition and scalar multiplication on $\mathcal{M}_{m\times n}(\mathbb{F})$:

$$\begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix} + \begin{bmatrix} b_{1,1} & \cdots & b_{1,n} \\ \vdots & \ddots & \vdots \\ b_{m,1} & \cdots & b_{m,n} \end{bmatrix} = \begin{bmatrix} a_{1,1}+b_{1,1} & \cdots & a_{1,n}+b_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1}+b_{m,1} & \cdots & a_{m,n}+b_{m,n} \end{bmatrix}$$

$$\alpha \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix} = \begin{bmatrix} \alpha a_{1,1} & \cdots & \alpha a_{1,n} \\ \vdots & \ddots & \vdots \\ \alpha a_{m,1} & \cdots & \alpha a_{m,n} \end{bmatrix}$$

for every $a_{i,j}, b_{i,j}, \alpha \in \mathbb{F}$, and this makes $\mathcal{M}_{m\times n}(\mathbb{F})$ a vector space over $\mathbb{F}$. The zero vector here is just the zero matrix $O_{m\times n}$ whose entries are all 0.

Exercise 1.6

Consider the set:

$$\mathbb{F}^\infty = \{(x_1, x_2, x_3, ...) : \text{ each } x_k \in \mathbb{F}\}$$

of all sequences of elements in $\mathbb{F}$. We can represent $(x_1, x_2, x_3, ...)$ concisely as $(x_k)_{k=1}^\infty$.

Define the operations $+ : \mathbb{F}^\infty \times \mathbb{F}^\infty \to \mathbb{F}^\infty$ and $\cdot : \mathbb{F} \times \mathbb{F}^\infty \to \mathbb{F}^\infty$ by:

$$(x_k)_{k=1}^\infty + (y_k)_{k=1}^\infty = (x_k + y_k)_{k=1}^\infty,$$

$$c \cdot (x_k)_{k=1}^\infty = (cx_k)_{k=1}^\infty,$$

for every $(x_k)_{k=1}^\infty, (y_k)_{k=1}^\infty \in \mathbb{F}^\infty$ and $c \in \mathbb{F}$. Show that $\mathbb{F}^\infty$ is a vector space over $\mathbb{F}$ under these operations.

Lecture 2 More examples and vector space properties

This covers part of ***Section 1.1*** *in the course notes.*

We now look at more examples of vector spaces as well as properties of vector spaces. For the first example, it is useful to revisit polynomials from MATH 135.

Definition 2.1 (Polynomial)

A **polynomial** with coefficients in $\mathbb{F}$ is an expression of the form:

$$p(x) = \sum_{j=0}^{n} a_j x^j = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

where $n \geq 0$, x is an indeterminate, and $a_0, a_1, ..., a_n \in \mathbb{F}$ are coefficients. If every $a_j = 0$, we write $p(x) = 0$ and call this the **zero polynomial**.

The **degree** of $p(x)$ is given by:

$$\deg(p(x)) = \begin{cases} \max\{j \geq 0 : a_j \neq 0\}, & \text{if } p(x) \neq 0 \\ -\infty, & \text{if } p(x) = 0 \end{cases}$$

with the convention that $-\infty$ is strictly less than any $n \geq 0$. For example:

$$\deg(2x - 1) = 1, \quad \deg(3x^3 - 5x^2 + 4) = 3, \quad \deg(2) = 0.$$

Example 2.2 (Space of polynomials)

Let

$$\mathcal{P}(\mathbb{F}) = \{a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n : a_0, a_1, ..., a_n \in \mathbb{F}, n \geq 0\}$$

be the set of all polynomials in the indeterminate x with coefficients in $\mathbb{F}$. Consider two polynomials:

$$p(x) = \sum_{j=0}^{n} a_j x^j = a_0 + a_1 x + \cdots + a_n x^n, \quad q(x) = \sum_{j=0}^{m} b_j x^j = b_0 + b_1 x + \cdots + b_m x^m$$

in $\mathcal{P}(\mathbb{F})$, where $n, m \geq 0$. We can let $N = \max\{n, m\}$ and define $a_i = 0$ for $i > n$ and $b_i = 0$ for $i > m$. This allows us to write:

$$p(x) = \sum_{j=0}^{N} a_j x^j = a_0 + a_1 x + \cdots + a_N x^N, \quad q(x) = \sum_{j=0}^{N} b_j x^j = b_1 + b_1 x + \cdots + b_N x^N$$

so that the last term for each polynomial is of the form cx^N. We can then set:

$$p(x) + q(x) = \sum_{j=0}^{N} (a_j + b_j) x^j = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_N + b_N)x^N.$$

This is how addition is defined on $\mathcal{P}(\mathbb{F})$. Basically, we add two polynomials by collecting like terms (the powers of x) to obtain another polynomial of the form $\sum_{j=0}^{N} c_j x^j$. For example:

$$(1 - 2x + 3x^2) + (-2 + 4x - x^2 + 7x^3) = -1 + 2x + 2x^2 + 7x^3.$$

Similarly if we let $c \in \mathbb{F}$, we define scalar multiplication on $\mathcal{P}(\mathbb{F})$ by:

$$cp(x) = \sum_{j=0}^{n} (ca_j)x^j.$$

It can be checked that the usual addition of polynomials and scalar multiplication by $\mathbb{F}$ on polynomials makes $\mathcal{P}(\mathbb{F})$ a vector space over $\mathbb{F}$ (convince yourself this is true, like with $\mathbb{F}^n$). The zero vector here is just the zero polynomial 0.

Example 2.3 (Space of polynomials of degree $\leq n$)

Let:

$$\mathcal{P}_n(\mathbb{F}) = \{p(x) \in \mathcal{P}(\mathbb{F}) : \deg(p(x)) \leq n\}.$$

In particular, this means:

$$\mathcal{P}_n(\mathbb{F}) = \{a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n : a_0, a_1, ..., a_n \in \mathbb{F}\}$$

(notice here that n is fixed before defining the set, unlike with $\mathcal{P}(\mathbb{F})$ where n is free to be any non-negative integer).

With the addition and scalar multiplication given in **Example 2.2**, one can check that $\mathcal{P}_n(\mathbb{F})$ is also a vector space over $\mathbb{F}$. We refer to this as the vector space of **polynomials of degree $\leq n$**.

Note that we often write $\mathcal{P}_0(\mathbb{F})$ as just $\mathbb{F}$.

Remark 2.4

Here is a **warning**: it is important *not* to think of polynomials as functions on $\mathbb{F}$. Rather, you should treat these as symbolic expressions which you can add and scalar multiply (like we defined above) as well as multiply.

One reason for this is there are situations in this course where we shall be plugging in more than just scalar values into polynomial expressions (notably, we will be doing this for square matrices when we do the *Cayley-Hamilton theorem* later in the course) so it is useful to view polynomials as symbolic.

There are also circumstances later in math where you could be working with polynomials over other fields in which a non-zero polynomial could be zero when you plug in *any* element of the field. For example, you may be familiar with the field $\mathbb{Z}/3\mathbb{Z}$ of integers modulo 3 from MATH 135. Over $\mathbb{Z}/3\mathbb{Z}$, the

polynomial $f(x) = x^3 - x$ is non-zero, but has that $f(0) = f(1) = f(2) = 0$.

We won't be dealing with fields other than $\mathbb{R}$ or $\mathbb{C}$ in this course though.

Example 2.5 (Continuous functions)

Consider the set:

$$\mathcal{C}([0,1]) = \{f : [0,1] \to \mathbb{R} : f \text{ is continuous on } [0,1]\}.$$

Given $f, g \in \mathcal{C}([0,1])$ and $\alpha \in \mathbb{R}$, we define addition and scalar multiplication by:

$$(f+g)(x) = f(x) + g(x) \text{ for all } x \in [0,1],$$
$$(\alpha f)(x) = \alpha f(x) \text{ for all } x \in [0,1].$$

Note that the sum of continuous functions is again continuous, so $f+g$ is in $\mathcal{C}([0,1])$. Similarly the product of two continuous functions is continuous and the constant function given by $h(x) = \alpha$ for all $x \in [0,1]$ is continuous, so $\alpha f = hf$ is in $\mathcal{C}([0,1])$. So these operations indeed make sense on the space $\mathcal{C}([0,1])$.

It can be shown, using properties of functions and $\mathbb{R}$, that $\mathcal{C}([0,1])$ is a vector space over $\mathbb{R}$. The zero vector is the constant zero function, which we can denote by 0.

Example 2.6 (Zero vector space)

There is also a trivial vector space, which we will see pops up quite frequently later in the course. Here we consider $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and the vector space $V = \{\vec{0}\}$ consisting of exactly one element $\vec{0}$ with the operations:

$$\vec{0} + \vec{0} = \vec{0}$$
$$\alpha \cdot \vec{0} = \vec{0} \ \ \forall \alpha \in \mathbb{F}$$

V is called the **zero vector space** over $\mathbb{F}$ and it contains only the zero vector. We will denote it by $\{\vec{0}\}$.

An important property of vector spaces is the following pair of cancellation laws.

Proposition 2.7 (Cancellation laws)

Let V be a vector space over $\mathbb{F}$, let $\vec{u}, \vec{v}, \vec{w} \in V$, and let $\alpha \in \mathbb{F}$.

(i) If $\vec{u} + \vec{v} = \vec{u} + \vec{w}$, then $\vec{v} = \vec{w}$.

(ii) If $\alpha\vec{u} = \alpha\vec{v}$ and $\alpha \neq 0$, then $\vec{u} = \vec{v}$.

Proof.

(i) Suppose first that $\vec{u}+\vec{v}=\vec{u}+\vec{w}$. We can set $\vec{y}=\vec{u}+\vec{v}=\vec{u}+\vec{w}$. Then $\vec{y}-\vec{u}$ has two expressions:

$$\vec{y}-\vec{u}=\vec{u}+\vec{v}-\vec{u},$$
$$\vec{y}-\vec{u}=\vec{u}+\vec{w}-\vec{u}.$$

Applying our vector space axioms, we have:

$$\vec{u}+\vec{v}-\vec{u}=\vec{u}-\vec{u}+\vec{v}=\vec{0}+\vec{v}=\vec{v}$$

(can you see which vector space axiom was used at each step of the chain of equalities above?) Similarly we have $\vec{u}+\vec{w}-\vec{u}=\vec{w}$, so we get $\vec{v}=\vec{w}$.

(ii) Now suppose $\alpha\vec{u}=\alpha\vec{v}$ and $\alpha\neq 0$. Since $\alpha\neq 0$, $\frac{1}{\alpha}$ is defined, and we have $\frac{1}{\alpha}(\alpha\vec{u})=\frac{1}{\alpha}(\alpha\vec{v})$. Now:

$$\frac{1}{\alpha}(\alpha\vec{u})=\frac{\alpha}{\alpha}\vec{u}=1\cdot\vec{u}=\vec{u}$$

and similarly $\frac{1}{\alpha}(\alpha\vec{v})=\vec{v}$, so $\vec{u}=\vec{v}$ and the proof is complete.

□

These laws, while simple, are what allow us to manipulate algebraic expressions involving vectors, letting us solve equations and prove more properties.

Exercise 2.8

Let V be any vector space over $\mathbb{F}$ and let $\vec{v},\vec{w}\in V$ be any vectors.

Show that there is a *unique* solution to the following vector equation:

$$\vec{v}+3\vec{x}=\vec{w}$$

where $\vec{x}$ is the variable to be solved for.

Exercise 2.9

Let V be a vector space over $\mathbb{F}$. Prove the following:

(i) $0\vec{v}=\vec{0}$ for all $\vec{v}\in V$.

(ii) $(-1)\vec{v}=-\vec{v}$ for all $\vec{v}\in V$.

(iii) $\alpha\vec{0}=\vec{0}$ for all $\alpha\in\mathbb{F}$.

Why do these identities make sense?

$\mathbb{F}^n$, $\mathcal{P}(\mathbb{F})$, $\mathcal{P}_n(\mathbb{F})$, $\mathcal{M}_{m\times n}(\mathbb{F})$, and $\mathcal{C}([0,1])$ are what we could call "mainstream" vector spaces. For us, this just means almost all of the examples we use throughout this course will be one of these. The following gives an example of an uncommon vector space, and illustrates the vector spaces can be found in some surprising places.

Exercise 2.10

Let $V = \mathbb{R}_{>0} = \{x \in \mathbb{R} : x > 0\}$ and $\mathbb{F} = \mathbb{R}$ with addition $\oplus$ and scalar multiplication $\odot$ given by:

$$\begin{aligned} x \oplus y &= x \cdot y \quad \text{for all } x, y \in V \\ \alpha \odot x &= x^{\alpha} \quad \text{for all } x \in V,\ \alpha \in \mathbb{R} \end{aligned}$$

Prove that V is a vector space over $\mathbb{F}$ under the above operations.